

① Spock's Dilemma



n planets in a system
k planets visited in allotted time
visit order does not matter

How many combinations of planets can he visit?

Let's start by assuming Spock is fascinated with one planet called Planet X.

Two possibilities:

1. Visit Planet X
2. Do not visit Planet X

We have two parameters: k and n
Let $C(n, k)$ = number of possible combinations

Possibility 1 - Visit Planet X

k-1 left to select

n-1 remaining to select from

$C(n-1, k-1)$ remaining combinations

Possibility 2 - Do not visit Planet X

k left to select

n-1 remaining to select from

$C(n-1, k)$ remaining combinations

Spock deduces:

$C(n, k) = (\text{\# of } k \text{ groups including Planet X})$
+ $(\text{\# of } k \text{ groups excluding Planet X})$

Therefore:

$C(n, k) = (\text{possibility 1}) + (\text{possibility 2})$

$C(n, k) = C(n-1, k-1) + C(n-1, k)$

Fascinating...

② Base cases

1. Visit them all? $k=n$

$$C(n, k) = C(k, k) = 1$$

One combination

2. Cannot visit any? $k=0$

$$C(n, 0) = 1$$

One combination that is the empty set.

3. Visit more than exists? $k>n$

Not realistic but we often need to consider all possibilities.

$C(n, k) = 0$ if and only if $k>n$

$$C(n, k) = \begin{cases} 1 & , k=0 \\ 1 & , k=n \\ 0 & , k>n \\ C(n-1, k-1) + C(n-1, k) & , 0 < k < n \end{cases}$$

$C(n, k)$

if $((k==n) \parallel (k==0))$

return 1

else if $(k>n)$

return 0

else

return $C(n-1, k-1) + C(n-1, k)$

③ $C(n, k)$
if $((k==n) \parallel (k==0))$
return 1
else if $(k>n)$
return 0
else
return $C(n-1, k-1) + C(n-1, k)$



Spock's challenge

1. Code the method and try to find the n and k limits
2. Draw the box trace for $C(9, 5)$
3. Draw the recursion tree for $C(9, 5)$